

Md Saif Ahmad

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🌐 Portfolio : saifahmad.github.io | 🐙 github.com/saifahmadgit | 🔗 linkedin.com/in/saif-ahmad

EDUCATION

Northwestern University – Evanston, Illinois	M.S in Robotics (Sept 2025 – Sep 2026(expected))
IIT Guwahati – Guwahati, India	M.Tech in Mechanical Engineering (July 2018 – May 2020)
Aligarh Muslim University – Aligarh, India	B.Tech in Mechanical Engineering (July 2013 – May 2017)

SKILLS

Robotics: Reinforcement Learning, VLA, Legged Locomotion, SLAM, Perception, Computer Vision, Path Planning, Navigation

Software: ROS2, C++, Python, C, MATLAB, Git, Linux, Bash, Embedded programming, Perforce, Docker

Machine Learning: Pytorch, Tensorflow, SKlearn, Deep Reinforcement learning, Generative AI

Simulation: Issac Sim, Issac Lab, Genesis, Gazebo, CoppeliaSim, Simulink, GT Suite

Hardware: Raspberry Pi, Arduino, NVIDIA Jetson

PROFESSIONAL EXPERIENCE

Modelling and Simulation Engineer - Daimler Trucks	October 2020 – July 2025
[MATLAB, Simulink, GT-Suite, Python, Pytorch, Tensorflow, Vector CANdb++, Perforce]	Bangalore, India

- Collaborated with the energy management team to develop fuel-cell/battery power-split strategies, improving range by 8%.
- Developed a battery fault-detection algorithm in PyTorch using an LSTM autoencoder for limited labeled fault data.
- Built EKF/UKF-based SOC estimators with the BMS team to correct drift from current sensor error, published in SAE.
- Designed a gain-scheduled PI + feedforward controller for a human driver pedal actuation model, improving velocity tracking.
- Led development and delivery of Hardware-in-the-Loop plant models for a new CAN architecture, managing a 3-engineer team.
- Built EV powertrain digital twins in Simulink/GT-Suite simulating range, performance, and thermal behavior at 95% accuracy.
- Validated ECU control logic using Simulink S-functions in closed-loop with a vehicle model for SIL/MIL testing and tuning.

PROJECTS

Sim-to-Real VLA for Contact-Rich Manipulation

[Pi 0.5, ACT, Issac Sim, ROS2, Python]	Mar 2026 – Present
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- Built an end-to-end sim-to-real pipeline, deploying fine-tuned Pi 0.5 on a Franka Panda.
- Bridging the sim-to-real gap across multiple generalist robot policies to enable automated Isaac Sim data to directly drive real-world deployment, eliminating the need for real-world data collection.

Quadruped Locomotion Reinforcement Learning Sim to Real on Unitree Go2

[Reinforcement Learning, PPO, Genesis, Python, Unitree Go2]	Jan 2026
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- Trained PPO-based reinforcement learning policies in Genesis for walking, jumping, crouching, and stair climbing in simulation.
- Achieved robust sim-to-real transfer for walking and staircase climbing.
- To close sim to real gap used domain randomization, simulated sensor noise and latency and learned per leg stiffness.

LeHome Challenge – ICRA 2026 Garment Manipulation Competition

[VLA, Diffusion Policy, Pi 0.5, Issac Sim, Python, PyTorch, LeRobot SO-ARM101]	Mar 2026 – Present
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- Deploying Vision-Language-Action (VLA) policies for robust deformable garment manipulation.
- Worked on developing automated data collection pipeline in Simulation using cuRobo.

EKF SLAM Pipeline for TurtleBot3 Burger in C++ from Scratch

[ROS 2, SLAM, C++]	Jan 2026
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- Programmed a complete ROS 2 pipeline in C++ for SLAM on a TurtleBot, from scratch.
- Developed a kinematics and control library with a simulation framework to enable sim-to-real deployment on TurtleBot3.

Franka Emika Panda Arm – Vision-Guided Pick and Place

[ROS 2, MoveIt 2, Intel RealSense, OpenCV, YOLO]	Dec 2025
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- Built a vision-guided manipulation pipeline for the Franka to detect, match, and autonomously grasp and place objects.
- Integrated Intel RealSense inputs, YOLO, and MoveIt via ROS 2 for autonomous grasp-and-place control.

Optimization of Bearing Parameters

[C++, Genetic Algorithms]	Jan 2020
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- Conducted multi-objective optimization of bearing design to maximize load capacity and minimize wear, published in ASME.
- Ensured the optimized design was robust to manufacturing tolerances by incorporating variability into the GA framework.